

Michael Pignatelli

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EDUCATION

University of Pennsylvania, School of Engineering and Applied Science

Philadelphia, PA

Bachelor of Science in Engineering in Computer Science and Cognitive Science (Dual Major)

Expected May 2027

- Minor in Italian Culture | GPA: 3.56/4 | Affiliations: Claude Builder Club, Wharton Data Science Group

Master of Science in Engineering in Computer and Information Science (Sub-matriculation)

Expected 2027

- Concentration in Artificial Intelligence | GPA: 3.85/4

Università di Bologna | *Exchange Student – Bologna Consortial Studies Program*

January 2026 – June 2026

- Coursework in NLP, Cognitive Neuroscience, and advanced Italian language & culture at the world's oldest university.

EXPERIENCE

University of Pennsylvania | *Head Teaching Assistant – Computer Science (CIS 1100)*

August 2024 – Present

- Lead and mentor 40+ undergraduate and graduate TAs for Penn's largest intro CS course (~300 students), overseeing recitations, office hours, and grading; maintained TA quality rating of 3.41/4 – well above CIS dept. average.
- Develop and deliver instruction on data structures, recursion, OOP, functional programming, and Pandas.
- Guide students from “Hello, World!” through advanced topics to a final movie recommendation engine project.
- Maintain course infrastructure (cis1100.com) with HTML/YAML/CSS, reducing site errors and ensuring clarity.
- Collaborate with faculty to update curriculum, author exams, and lead weekly staff meetings.

Bentley Systems | *Software Development Engineering Intern*

June 2026 – August 2026

- Integrated Bentley Copilot, an AI assistant, into the SACS Offshore and iTwin Studio platforms to enable AI-driven iModel report generation and analysis.
- Collaborated with a fellow intern to build MCP (Model Context Protocol) server integrations connecting SACS Offshore DB and other analysis tools to Bentley Copilot.
- Implemented Azure AD authentication required by the Bentley Copilot library to support secure AI tool integration.
- Worked primarily in TS/JavaScript/React, with additional work in Python.

Ava Labs | *Software Engineering Intern – Platform Team*

June 2025 – August 2025

- Implemented and productionized 4 gRPC APIs (+7.5k LoC) in the AvalancheGo blockchain client, replacing legacy JSON-RPC and enabling strongly-typed communication for all node operators. Reduced parsing errors and improved type safety, delivering ~50% lower latency vs REST for ~1,800 node operators.
- Wrote end-to-end and integration tests (Ginkgo) to validate gRPC API adoption and ensure cross-version stability.
- Authored incremental deployment plan adopted for long-term rollout.
- Collaborated with Data Tooling and Cloudflare teams to align gRPC rollout with multi-team infrastructure standards.

Scale AI (via Outlier AI) | *Prompt Engineer (Contract)*

May 2024 – August 2024

- Engineered and validated 500+ prompts for LLM fine-tuning on math reasoning and algorithmic coding tasks, improving model accuracy and dataset consistency, across diverse math and programming benchmarks.
- Reviewed peer prompts and enforced quality standards, ensuring consistency across datasets and difficulty levels.

SELECTED PROJECTS

Tabby — Built full-stack bill-splitting app (FastAPI, MongoDB, React) using Google Gemini API; implemented tax/tip distribution, Venmo integration, payment tracking; demoed to 500+ peers at PennApps Hackathon under 36-hour deadline.

PennOS — Built a Unix-like operating system in C for CIS 5480, implementing processes, scheduling, signals, a shell, and a file system; primary contributor to the PennFAT file system design and implementation.

Lifeline — Designed and built an accessible healthcare management web app (React, TS, TailwindCSS, Gemini API) for CIS 4120 (HCI); led research, prototyping, and heuristic evaluation, and implemented the core UI and AI integration [GitHub].

Pedestrian Buddies — Built embedded CV pipeline (TensorFlow Lite, Keras) for real-time crossing guard gesture detection; optimized inference for hardware-constrained systems, deployed on Arduino using TensorFlow Lite runtime [GitHub].

Housing & Education Analysis — Analyzed 1.6M+ U.S. housing & education records, including state-level educational spending; applied ridge regression, bootstrapping, and random forests to study spending–price relationships [Report].

Guess Who? — Developed Java Swing multiplayer game with client-server architecture and robust state sync [GitHub].

LEADERSHIP & TECHNICAL INVOLVEMENT

Penn High Powered Rocketry | *Active Recovery Engineering Lead*

September 2023 – Present

- Lead recovery system design for L3 rocket, implementing dual parachute deployment. Simulated flight dynamics (<5% error vs actual), contributing to successful 12,000 ft launch at Spaceport America Cup/IREC (120+ teams).
- Conducted 3,600 ft field tests validating parachute energetics and deployment timing under live launch conditions.
- Integrated telemetry and comms testing with recovery hardware to ensure redundancy and reliable system performance.

Penn Engineers Without Borders | *Ramp Committee Engineer*

January 2025 – December 2025

- Designed ADA-compliant ramp in AutoCAD for Grace City Church, ensuring accessibility for 200+ weekly visitors.

SKILLS

Programming & Frontend: Python, Go, Java, C/C++, OCaml, SQL, Swift, JS/TS, React, SwiftUI, HTML/CSS, Figma
Backend/Infra: Docker/Containerization, Git/GitHub (Actions), CI/CD, Testing (JUnit, Ginkgo, pytest), Linux, gRPC/Protobuf, REST, FastAPI, MongoDB, Cloudflare

ML/Data Science: NLP, RAG, MCP, Fine-Tuning, Prompt Engineering, TensorFlow/PyTorch/Keras, Pandas/NumPy

APIs/Tools: Google Gemini, OpenAI, Anthropic, Jupyter, XML/JSON